

BIOL90042 tutorial 1

These questions will help you to revise the introduction to R/RStudio content from lecture 1. You may work independently or in groups. Use RStudio to help you, and if you have any questions please ask a tutor.

1. Name the four panels of RStudio, and briefly describe what each one is used for
2. What would be the result of evaluating the following in R?

- a. $200 / 10$
- b. $4 \wedge 2$
- c. 4^2
- d. $0.1 * 6$
- e. $5/0$
- f. $1 + 2 + 3 + 4$
- g. $2 + 3 * 4$
- h. $(2 + 3) * 4$
- i. $10 + 6 \ 0$

3. Write R code to solve the following mathematical problems

- a. $100 + 5$
- b. $0.2 \div 0.4$
- c. $2^2 + 3^2 + 4^2$
- d. $\frac{(15+2)^3}{100}$

The following question asks you about file paths. Please answer them according to the operating system of your computer. If there are people around you with a different operating system, try comparing your answers!

4. I have a file called 'tutorial_1.R' inside of a folder called 'BIOL90042_tutorials', which is located on my desktop.
 - a. Write a **relative** path to 'tutorial_1.R', from the desktop

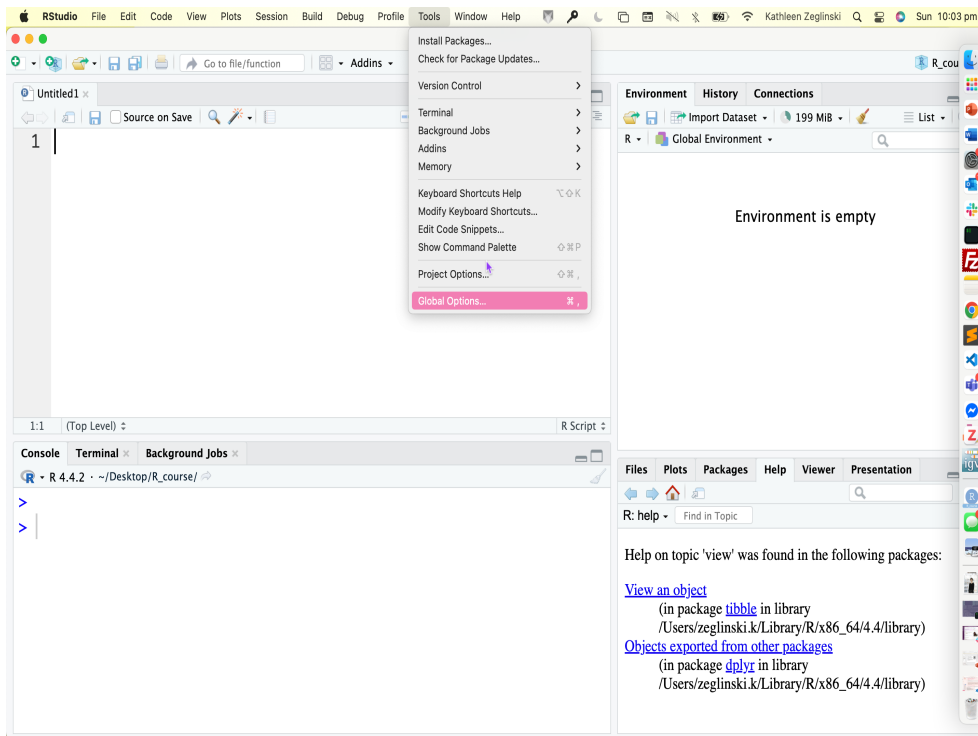
- b. Write an **absolute** path to 'tutorial_1.R'
- c. Why did I name my file 'tutorial_1.R' instead of 'tutorial 1.R'?
- d. If I told you I'd made a mistake, and the folder was actually called 'Biol90042_tutorials', would the paths you wrote for parts a and b still work? Why/why not?

Hint: If you're struggling, remember that you can use the gear button in the RStudio files panel to copy the file path. You could try creating these files/folders on your own computer use the gear button to copy the path to them.

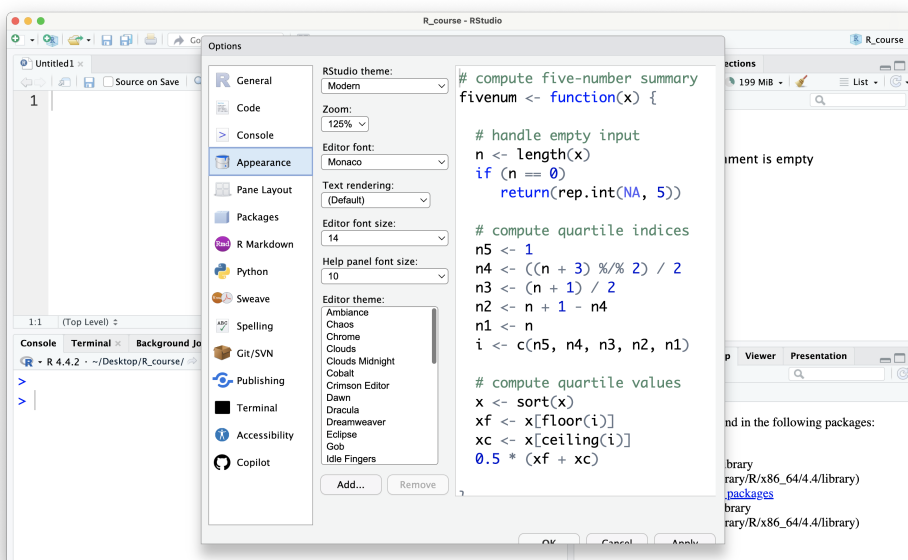
5. Challenge: I want to use R to calculate the volume of a sphere (recall that the formula for this is $\frac{4}{3}\pi r^3$, where r is the radius of the sphere). My sphere has a radius of 3 cm.
 - a. Using the approximation $\pi \approx 3.14$, write some R code to calculate the volume of this sphere.
 - b. Now, re-write the code using pi instead of 3.14. Are the answers the same? Why/why not?
 - c. Which line of code, the one from part a or part b do you think would be the best way to calculate the volume of a sphere in practice? Why?
 - d. pi is what is known as a *constant*, it's a special name for a value that won't ever change. The opposite of a constant is a *variable*. In the formula above, what is the variable?
 - e. In R, you can assign a value to a variable using the assignment operator <- (type < and then - on your keyboard). For example, to assign the value of 2 to a variable called x, you would write x <- 2. Use this information to re-write the code from part b to use a variable.
 - f. What do you think is the advantage of using variables in R?

Just for fun: customising your RStudio!

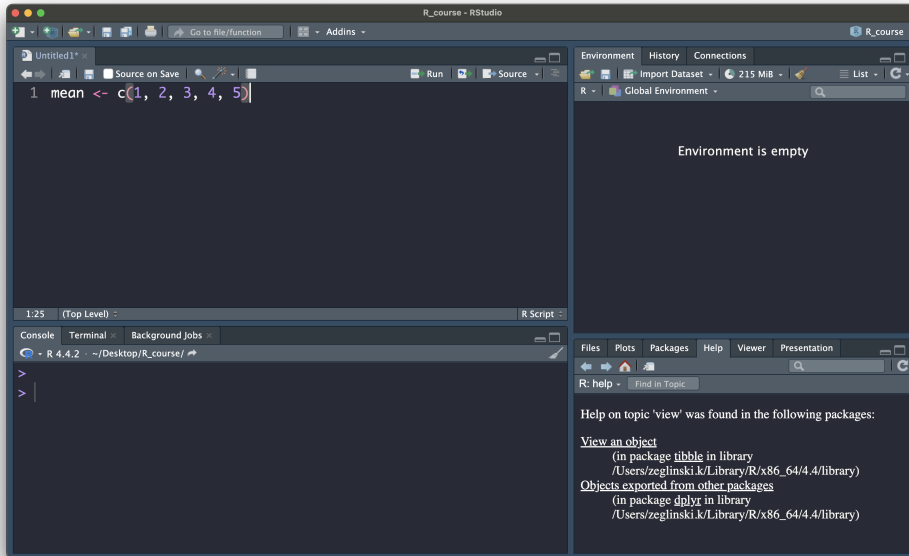
RStudio is highly customisable, and you can change the appearance of the interface to suit your preferences. Try changing the theme, font size, or layout of RStudio to make it more comfortable for you to work in. You can find these options in the RStudio settings (In the menu bar go to Tools > Global Options):



For example, you can change the theme in the ‘Appearance’ section:



Here’s the ‘dracula’ theme:



Find a theme that you like and customise your RStudio!

There are also a bunch more themes made by others available for download online: <https://github.com/max-alletsee/rstudio-themes>, or you can even make your own.